

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

MERN STACK

Practical – I

(Environment Setup and JavaScript Basics)

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Class: TYIT	Batch: 2
Date of Assignment: 30-06-26	Date/Time of Submission: 30-06-26

a. Install Node.js and Visual Studio Code, and verify installation.

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● PS C:\Users\info\OneDrive\Desktop\mern prac> node -v
v24.17.0
● PS C:\Users\info\OneDrive\Desktop\mern prac> npm -v
Debugger attached.
11.13.0
Waiting for the debugger to disconnect...
○ PS C:\Users\info\OneDrive\Desktop\mern prac> 

```

b. Create a simple Node.js file and run it using the terminal.

```

JS prac.js  JS pracB.js  X
JS pracB.js
1 console.log("Hello, Welcome to 1st Practical of MERN ");
2 console.log("khushiiiiiii");

```

OUTPUT:-

```

● PS C:\Users\info\OneDrive\Desktop\mern prac> node pracB.js
Debugger attached.
Hello, Welcome to 1st Practical of MERN
khushiiiiiii
Waiting for the debugger to disconnect...

```

c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

```

JS prac.js  JS pracB.js  JS pracC.js  X
JS pracC.js > ...
1 console.log("Welcome to Node.js!");
2 let num1 = 10;
3 let num2 = 20;
4 console.log("First number: ", num1 );
5 console.log("second number: ", num1 );
6 console.log("Addition : ", num1 + num2 );
7 console.log("sub : ", num1 - num2 );
8 console.log("mul : ", num1 * num2 );
9 console.log("div : ", num1 / num2 );
10 console.log("Khushiiiiiii")

```

OUTPUT:-

```

PS C:\Users\info\OneDrive\Desktop\mern prac> node pracC.js
Debugger attached.
Welcome to Node.js!
First number: 10
second number: 10
Addition : 30
sub : -10
mul : 200
div : 0.5
Khushiiii

```

d. Create and run a simple program using variables and functions

```

JS pracJs JS pracBjs JS pracCjs JS pracDjs JS pracEjs
JS pracDjs > ...
1 let num1 = 20 ;
2 let num2 = 10 ;
3 //function to add two number
4 function add(a,b) {
5     return a + b;
6 }
7 //function to multiply two number
8 function mul(a,b) {
9     return a * b;
10 }
11 // Display the values
12 console.log("First number : ", num1);
13 console.log("second number : ", num2);
14 // call the function
15 console.log("addition = ", add(num1 , num2));
16 console.log("Multiplication = ", mul(num1 , num2));
17 console.log("Khushiiii")

```

OUTPUT:-

```

PS C:\Users\info\OneDrive\Desktop\mern prac> node pracD.js
Debugger attached.
First number : 20
second number : 10
addition = 30
Multiplication = 200
Khushiiii

```

e. Demonstrate use of operators and control statements in JavaScript

```

JS pracJs JS pracBjs JS pracCjs JS pracDjs JS pracEjs
JS pracEjs > ...
1 let a = 20 ;
2 let b = 40 ;
3 console.log("\nComparison Operators");
4 console.log ("a > b : ", a > b);
5 console.log ("a < b : ", a < b);
6 console.log ("a == b : ", a == b);
7 console.log ("a != b : ", a != b);
8 console.log("\nIf- Else Statement ");
9 if (a > b) {
10     console.log("a is greater than b ");
11 } else {
12     console.log("b is greater than or equal to a ");
13 }
14 // Switch Statement
15 console.log("\nSwitch Statement ");
16 let day = 2;
17 switch (day){
18     case 1 :
19         console.log("Monday");
20         break;
21     case 2 :
22         console.log("Tuesday");
23         break;
24     case 3 :
25         console.log("Wednesday");
26         break;
27     default :
28         console.log("Invalid day");
29 }
30 console.log("Khushiiii");
31

```

OUTPUT:-

```

PS C:\Users\info\OneDrive\Desktop\mern prac> node pracE.js
Debugger attached.

Comparison Operators
a > b : false
a < b : true
a == b : false
a != b : true

If- Else Statement
b is greater than or equal to a

Switch Statement
Tuesday
Khushiiiiii

```

f. Write programs using loops and template literals.

```

JS prac.js  JS pracB.js  JS pracC.js  JS pracD.js  JS
JS pracF.js > ...
1  let number = 8 ;
2  console.log( `Multipliation Table of $ {number}` );
3  for (let i = 1 ; i <= 10 ; i++)
4  {
5      console.log(`$ {number} x $ {i} = $ { number * i }`);
6  }
7  }
8  console.log("Khushiiiiii");

```

OUTPUT:-

```

PS C:\Users\info\OneDrive\Desktop\mern prac> node pracF.js
Debugger attached.
Multipliation Table of $ {number}
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
8 x $ {i} = $ { number * i }
Khushiiiiii

```

Student_Result_Management_System2

Aim: To install and configure the Node.js development environment and develop a simple Student Result Management System using basic JavaScript concepts.

Problem Statement: A school wants to develop a simple application using Node.js that calculates and displays a student's result.

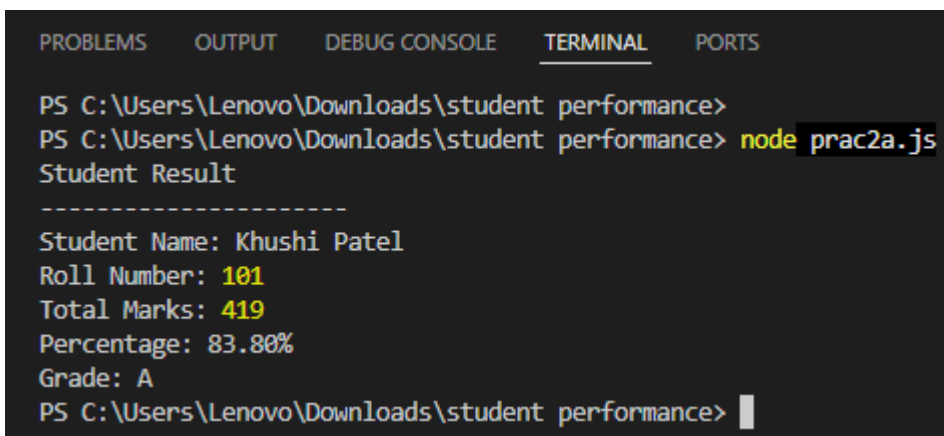
CODE (studentResult.js)

```

// Student Result Management System
let studentName = "Khushi Patel";
let rollNumber = 101;
let marks = [78, 85, 92, 88, 76];
let totalMarks = 0;
for (let mark of marks) {
    totalMarks += mark;
}
let percentage = totalMarks / marks.length;
let grade;

```

```
if (percentage >= 90) {  
    grade = "A+";  
} else if (percentage >= 80) {  
    grade = "A";  
} else if (percentage >= 70) {  
    grade = "B";  
} else if (percentage >= 60) {  
    grade = "C";  
} else {  
    grade = "F";  
}  
console.log("Student Result");  
console.log("-----");  
console.log("Student Name:", studentName);  
console.log("Roll Number:", rollNumber);  
console.log("Total Marks:", totalMarks);  
console.log("Percentage:", percentage.toFixed(2) + "%");  
console.log("Grade:", grade);
```

OUTPUT:-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\Lenovo\Downloads\student performance>  
PS C:\Users\Lenovo\Downloads\student performance> node prac2a.js  
Student Result  
-----  
Student Name: Khushi Patel  
Roll Number: 101  
Total Marks: 419  
Percentage: 83.80%  
Grade: A  
PS C:\Users\Lenovo\Downloads\student performance> |
```